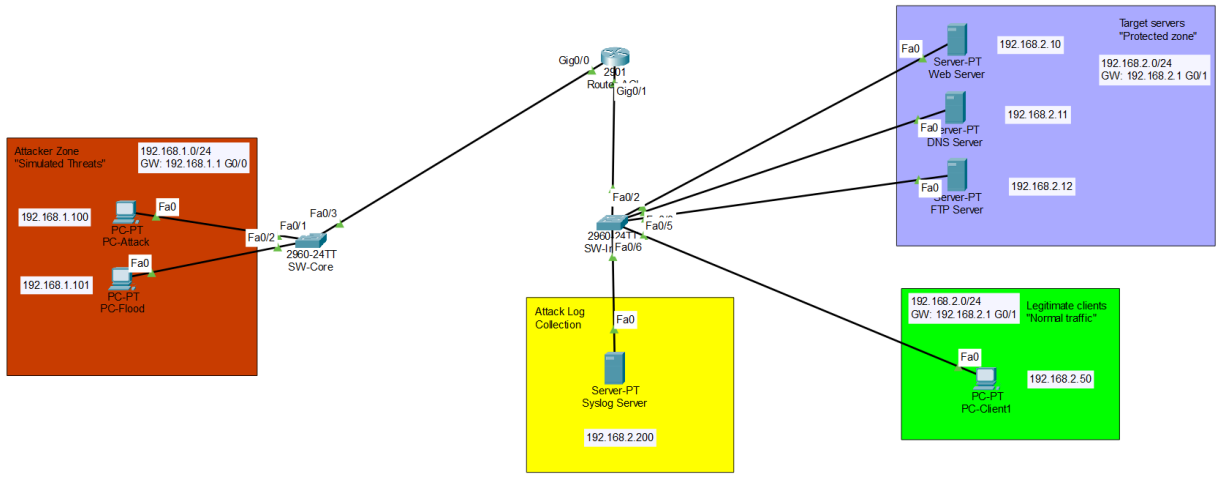
Lab Summary

Component	Device	Configuration	Status
DoS Mitigation ACL	Router-ACL	GigabitEthernet 0/0/0 and 0/0/1	Enabled
Port-Scanning ACL	Router-ACL	Extended ACL - ANTI-DOS	Applied
ACL Logging (log keyword)	Router-ACL	-	Unsupported in PT
Syslog Logging	Router-ACL	`logging host 192.168.2.200`	Working
Port Security	SW-Core Fa0/2	Maximum 1 MAC · Violation: Shutdown	Active
ICMP / Port 22 Traffic	PC-Attack	The ACL blocks the attacker	Blocked

Network Topology

- Full Network Topology - Cisco Packet Tracer



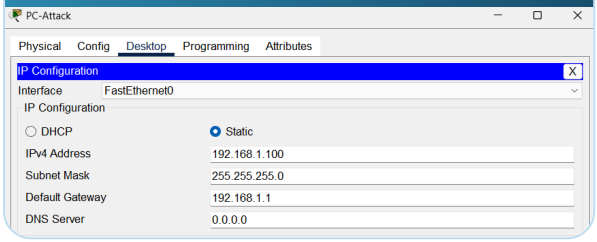
NOTE **Topology Overview**

The topology divides the network into three zones: the **attacker zone** (red), which contains PC-Attack and PC-Flood on `192.168.1.0/24` and connects to SW-Core and then Router-ACL; the **protected zone** (purple), which contains the Web, DNS, and FTP servers on `192.168.2.0/24`; and a **Syslog Server** (yellow) at `192.168.2.200` that collects logs. **PC-Client1** (green) represents a legitimate user. Router-ACL is the main segmentation point between the zones.

IP Configuration for All Devices

Static IP configuration for each device before starting the implementation:

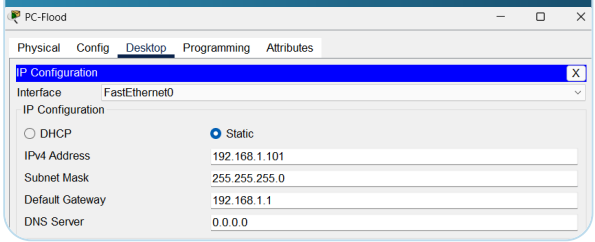
PC-Attack - Attacker Host



NOTE **IP Configuration - PC-Attack**

IP settings for **PC-Attack**. The device uses IP address `192.168.1.100`, subnet mask `255.255.255.0`, and default gateway `192.168.1.1`, which is interface Gi0/0/0 on the router. This host is the main source used to test the ACL and port-scanning behavior.

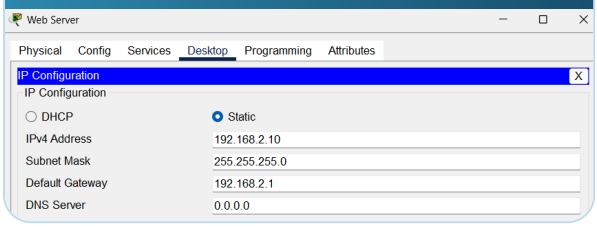
PC-Flood - Flood Host



NOTE **IP Configuration - PC-Flood**

IP settings for **PC-Flood**, which is used for MAC flooding attacks. Its IP address is `192.168.1.101` with the same subnet mask and default gateway. This host is connected to SW-Core on interface `Fa0/2`, which is the port secured by Port Security.

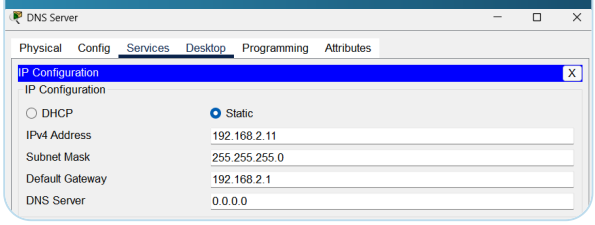
• Web Server



NOTE **IP Configuration - Web Server**

IP settings for the **Web Server** in the protected zone. The server uses IP address `192.168.2.10` and default gateway `192.168.2.1`, which is interface Gi0/0/1 on the router. The ACL applied on the router is intended to protect this server and the rest of the protected subnet from attacker traffic.

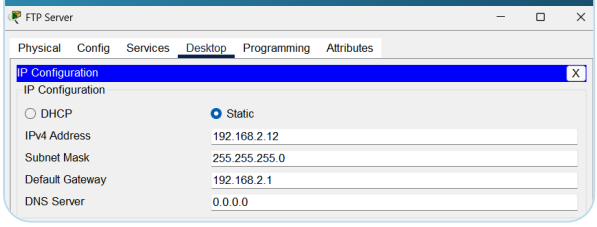
• DNS Server



NOTE **IP Configuration - DNS Server**

IP settings for the **DNS Server** in the protected zone. The server uses IP address `192.168.2.11` with the same gateway `192.168.2.1`. It is a potential target for the port-scanning attempts launched by PC-Attack, so it is protected by the ANTI-DOS ACL rules.

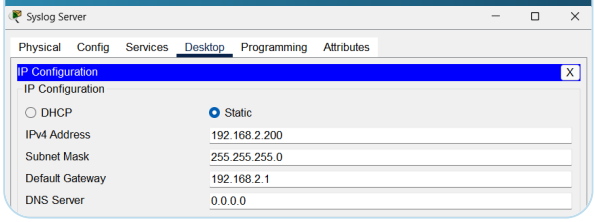
• FTP Server



NOTE **IP Configuration - FTP Server**

IP settings for the **FTP Server** in the protected zone. The server uses IP address `192.168.2.12`. It is directly affected by connection attempts to **port 22**, which are blocked by the ANTI-DOS ACL rule `deny tcp any any eq 22`.

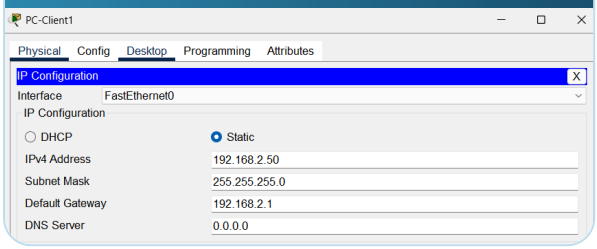
• Syslog Server



NOTE **IP Configuration - Syslog Server**

IP settings for the **Syslog Server**, which receives and stores log messages. The server uses IP address `192.168.2.200`. This is the same address configured on the router with `logging host 192.168.2.200` to receive Syslog messages.

- PC-Client1 - Legitimate User



NOTE **IP Configuration - PC-Client1**

IP settings for **PC-Client1**, which represents a legitimate user inside the protected zone. The device uses IP address `192.168.2.50` and default gateway `192.168.2.1`. This host is not affected by the ACL because the rule `permit ip any any` allows legitimate traffic to pass.

Router Interface Configuration

Enter global configuration mode and configure the interfaces on Router-ACL:

- GigabitEthernet 0/0/0 Interface

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router-ACL
Router-ACL(config)#interface GigabitEthernet0/0/0
Router-ACL(config-if)#ip address 192.168.1.1 255.255.255.0
Router-ACL(config-if)#no shutdown
```

NOTE **Configure Router Interface Gi0/0/0**

Commands used to configure the router from scratch: `en` to enter privileged EXEC mode, `config t` for global configuration mode, and a hostname change to `Router-ACL`. After that, interface `GigabitEthernet0/0/0` is assigned IP address `192.168.1.1` with subnet mask `255.255.255.0`, followed by `no shutdown` to enable the interface.

- GigabitEthernet 0/0/1 Interface

```
Router-ACL(config)#interface GigabitEthernet0/0/1
Router-ACL(config-if)#ip address 192.168.2.1 255.255.255.0
Router-ACL(config-if)#no shutdown
```

NOTE **Configure Router Interface Gi0/0/1**

Configuration of the second router interface, `GigabitEthernet0/0/1`, which connects to the protected zone. It is assigned IP address `192.168.2.1` with subnet mask `255.255.255.0`, followed by `no shutdown`. This interface acts as the default gateway for the servers and the Syslog Server in the protected network.

Create and Apply the ANTI-DOS ACL

- Create the ANTI-DOS Extended ACL

```
Router-ACL(config)#ip access-list extended ANTI-DOS
Router-ACL(config-ext-nacl)#deny icmp 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
Router-ACL(config-ext-nacl)#deny tcp any any eq 22
Router-ACL(config-ext-nacl)#deny tcp any any eq 23
Router-ACL(config-ext-nacl)#deny tcp any any eq 3389
Router-ACL(config-ext-nacl)#permit ip any any
Router-ACL(config-ext-nacl)#ex
```

NOTE **Create the ANTI-DOS Extended ACL**

The extended ACL named `ANTI-DOS` is created with `ip access-list extended ANTI-DOS`. The applied rules are `deny icmp 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255` to block ICMP from the attacker subnet to the protected subnet, `deny tcp any any eq 22` to block SSH, `deny tcp any any eq 23` to block Telnet, `deny tcp any any eq 3389` to block RDP, and `permit ip any any` to allow all remaining traffic.

• Apply the ACL and Configure Syslog

```
Router-ACL(config)#interface gigabitEthernet0/0/0
Router-ACL(config-if)#ip access-group ANTI-DOS in
Router-ACL(config-if)#ex
Router-ACL(config)#logging host 192.168.2.200
Router-ACL(config)#logging trap warnings
                                ^
% Invalid input detected at '^' marker.

Router-ACL(config)#logging trap 4
                                ^
% Invalid input detected at '^' marker.

Router-ACL(config)#service timestamps log datetime msec
Router-ACL(config)#
```

NOTE **Apply the ACL and Configure Syslog**

The ACL is applied inbound on the interface with `ip access-group ANTI-DOS in` to block traffic entering from the attacker network. Syslog is then configured with `logging host 192.168.2.200`, but Packet Tracer returns **Invalid input detected** when `logging trap warnings` is used. The workaround is `service timestamps log datetime msec`, which adds timestamps to the generated messages.

⚠️ **ACL Logging Limits in Packet Tracer**

The commands `logging trap warnings` and `logging trap 4` are not supported in Packet Tracer and return *Invalid input detected*. Syslog is used instead of the ACL `log` keyword.

Syslog Configuration and Testing

Interface Bounce - Generate Syslog Events

```
Router-ACL#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router-ACL(config)#interface g0/0/0
Router-ACL(config-if)#shutdown

Router-ACL(config-if)#no shutdown

Router-ACL(config-if)#end
*Mar 01, 00:38:13.3838: %LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to
administratively down
*Mar 01, 00:38:13.3838: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0,
changed state to down
*Mar 01, 00:38:13.3838: %LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up
*Mar 01, 00:38:13.3838: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0,
changed state to up
Router-ACL#
*Mar 01, 00:38:15.3838: SYS-5-CONFIG_I: Configured from console by console
Router-ACL#
```

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NOTE **Trigger Syslog Events with an Interface Bounce**

To test Syslog, an **interface bounce** is performed on `Gi0/0/0` by issuing `shutdown` followed by `no shutdown`. This generates Syslog messages that report interface-state changes such as `%LINK-5-CHANGED` and `%LINEPROTO-5-UPDOWN`. The result proves that the Syslog Server receives events successfully as a practical substitute for the unsupported ACL `log` keyword.

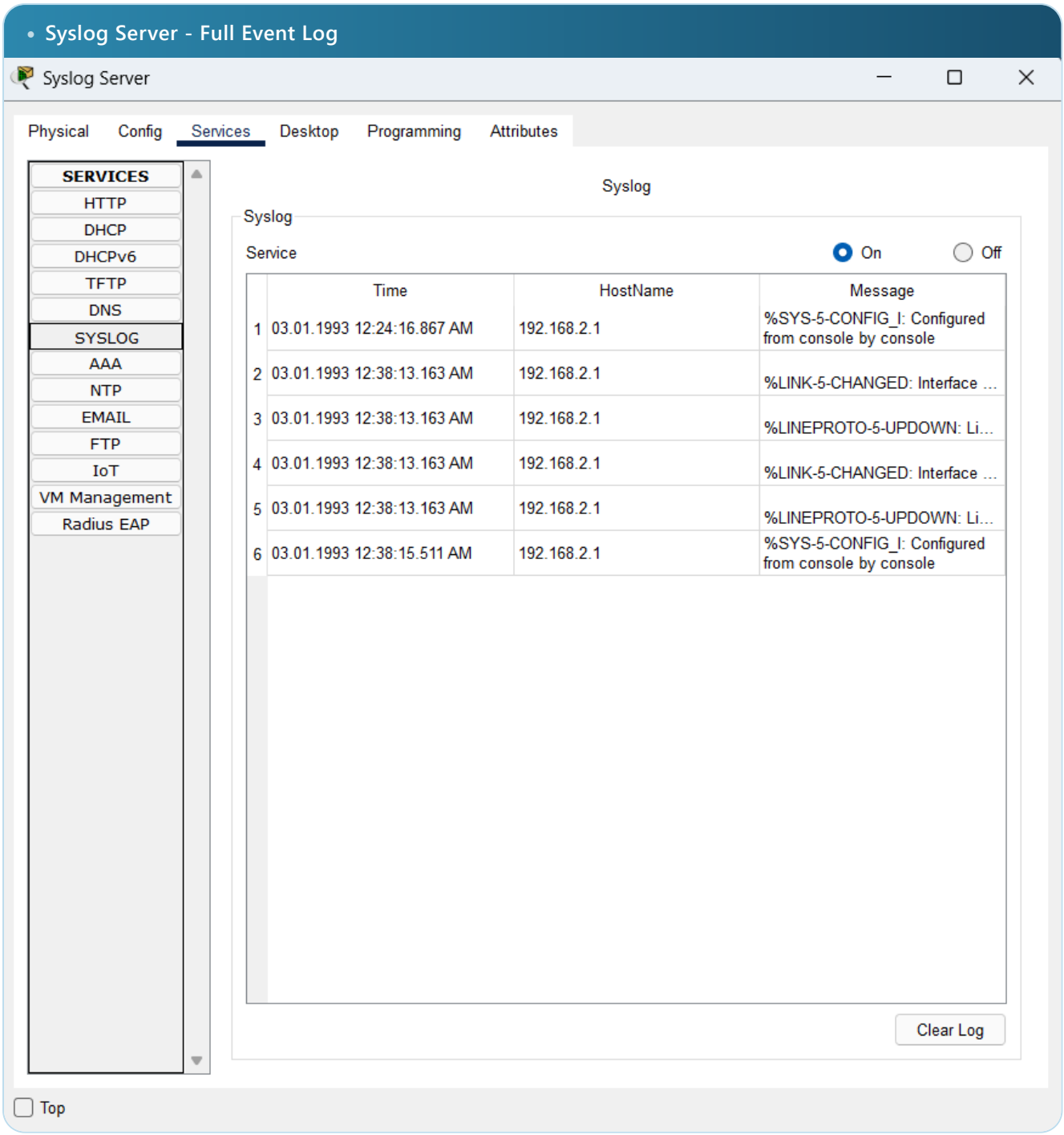
Syslog Command Summary

Bash

```bash
configure terminal
interface g0/0/0
shutdown
no shutdown
end
```

NOTE **Syslog Command Summary**

A short command sequence used to generate Syslog events: `configure terminal` -> `interface g0/0/0` -> `shutdown` -> `no shutdown` -> `end`. This sequence creates interface up/down events that appear on the Syslog Server as an alternative to the unsupported ACL logging feature.



NOTE **Syslog Server Event Log**

The **Syslog Server** displays all recorded events. The log contains `LINK-5-CHANGED` and `LINEPROTO-5-UPDOWN` messages for interface status changes, as well as `SYS-5-CONFIG_I` messages whenever the configuration is modified. This confirms that the Syslog Server is operating correctly and receiving events from the router.

Syslog Is Working Correctly

Syslog messages appear successfully on the server, which confirms that event logging works as an effective alternative to the unsupported ACL `log` keyword in Packet Tracer.

Port Security - SW-Core Fa0/2

Steps used to protect interface Fa0/2 from MAC flooding attacks:

1. **Enable Port Security**
   The command `switchport port-security` enables protection on the interface.

2. **Set Maximum MAC = 1**
   The command `switchport port-security maximum 1` prevents more than one MAC address from using the port.

3. **Violation Action: Shut Down the Port**
   The command `switchport port-security violation shutdown` automatically disables the port when a violation occurs.

4. **Sticky MAC Learning**
   The command `switchport port-security mac-address sticky` learns the legitimate MAC address and stores it in the running configuration.

- Configure Port Security on SW-Core Fa0/2

SW-Core

Physical Config CLI Attributes

IOS Command Line Interface

```
SW-Core>
SW-Core>enable
SW-Core#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW-Core(config)#interface fa0/2
SW-Core(config-if)#switchport port-security maximum 1
SW-Core(config-if)#switchport port-security violation shutdown
SW-Core(config-if)#end
SW-Core#
%SYS-5-CONFIG_I: Configured from console by console

SW-Core#
SW-Core#show port-security interface fa0/2
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 0009.7CD7.B961:1
Security Violation Count   : 0

SW-Core#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
               (Count)       (Count)        (Count)
--------------------------------------------------------------------
        Fa0/1        3          1                 0         Restrict
        Fa0/2        1          1                 0         Shutdown
        Fa0/3        3          1                 0         Restrict
        Fa0/4        3          0                 0         Restrict
        Fa0/5        3          0                 0         Restrict
        Fa0/6        3          0                 0         Restrict
        Fa0/7        3          0                 0         Restrict
        Fa0/8        3          0                 0         Restrict
        Fa0/9        3          0                 0         Restrict
       Fa0/10        3          0                 0         Restrict
---------------------------------------------------------------------
SW-Core#
```

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NOTE **Configure Port Security on SW-Core Fa0/2**

Port Security is configured on **SW-Core** to protect interface `Fa0/2`, which is connected to PC-Flood. The commands used are `switchport mode access` to set access mode, `switchport port-security` to enable protection, `switchport port-security maximum 1` to allow only one MAC address, `switchport port-security violation shutdown` to shut the port down on violation, and `switchport port-security mac-address sticky` to learn the legitimate MAC address automatically.

• Verification - Secure-Up State

SW-Core

Physical Config CLI Attributes

IOS Command Line Interface

```
Violation Mode              : Shutdown
Aging Time                  : 0 mins
Aging Type                  : Absolute
SecureStatic Address Aging  : Disabled
Maximum MAC Addresses       : 1
Total MAC Addresses         : 1
Configured MAC Addresses    : 0
Sticky MAC Addresses        : 1
Last Source Address:Vlan    : 0009.7CD7.B961:1
Security Violation Count    : 0

SW-Core#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
                (Count)       (Count)        (Count)
--------------------------------------------------------------------
        Fa0/1        3           1                 0         Restrict
        Fa0/2        1           1                 0         Shutdown
        Fa0/3        3           1                 0         Restrict
        Fa0/4        3           0                 0         Restrict
        Fa0/5        3           0                 0         Restrict
        Fa0/6        3           0                 0         Restrict
        Fa0/7        3           0                 0         Restrict
        Fa0/8        3           0                 0         Restrict
        Fa0/9        3           0                 0         Restrict
       Fa0/10        3           0                 0         Restrict
--------------------------------------------------------------------
SW-Core#show port-security interface fa0/2
Port Security               : Enabled
Port Status                 : Secure-up
Violation Mode              : Shutdown
Aging Time                  : 0 mins
Aging Type                  : Absolute
SecureStatic Address Aging  : Disabled
Maximum MAC Addresses       : 1
Total MAC Addresses         : 1
Configured MAC Addresses    : 0
Sticky MAC Addresses        : 1
Last Source Address:Vlan    : 0009.7CD7.B961:1
Security Violation Count    : 0

SW-Core#
SW-Core#
```

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NOTE **Verify Port Security - Secure-Up State**

The output of `show port-security interface Fa0/2` confirms that the configuration is correct: Port Security is **Enabled**, the port status is **Secure-up**, the violation mode is **Shutdown**, the maximum number of MAC addresses is **1**, and the number of learned sticky MAC addresses is **1**. Everything is operating as expected.

• Security Violation - Err-Disabled State

SW-Core

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

SW-Core#show port-security interface fa0/2
Port Security               : Enabled
Port Status                 : Secure-shutdown
Violation Mode              : Shutdown
Aging Time                  : 0 mins
Aging Type                  : Absolute
SecureStatic Address Aging  : Disabled
Maximum MAC Addresses       : 1
Total MAC Addresses         : 1
Configured MAC Addresses    : 0
Sticky MAC Addresses        : 1
Last Source Address:Vlan    : 0001.63A4.9DDE:1
Security Violation Count    : 1

SW-Core#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
                (Count)       (Count)        (Count)
--------------------------------------------------------------------
        Fa0/1        3           1                 0         Restrict
        Fa0/2        1           1                 1         Shutdown
        Fa0/3        3           1                 0         Restrict
        Fa0/4        3           0                 0         Restrict
        Fa0/5        3           0                 0         Restrict
        Fa0/6        3           0                 0         Restrict
        Fa0/7        3           0                 0         Restrict
        Fa0/8        3           0                 0         Restrict
        Fa0/9        3           0                 0         Restrict
       Fa0/10        3           0                 0         Restrict
--------------------------------------------------------------------
SW-Core#show interfaces status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    1          auto    auto  10/100BaseTX
Fa0/2                        err-disabled 1          auto    auto  10/100BaseTX
Fa0/3                        connected    1          auto    auto  10/100BaseTX
Fa0/4                        notconnect   1          auto    auto  10/100BaseTX
Fa0/5                        notconnect   1          auto    auto  10/100BaseTX
Fa0/6                        notconnect   1          auto    auto  10/100BaseTX
Fa0/7                        notconnect   1          auto    auto  10/100BaseTX
Fa0/8                        notconnect   1          auto    auto  10/100BaseTX
Fa0/9                        notconnect   1          auto    auto  10/100BaseTX
```

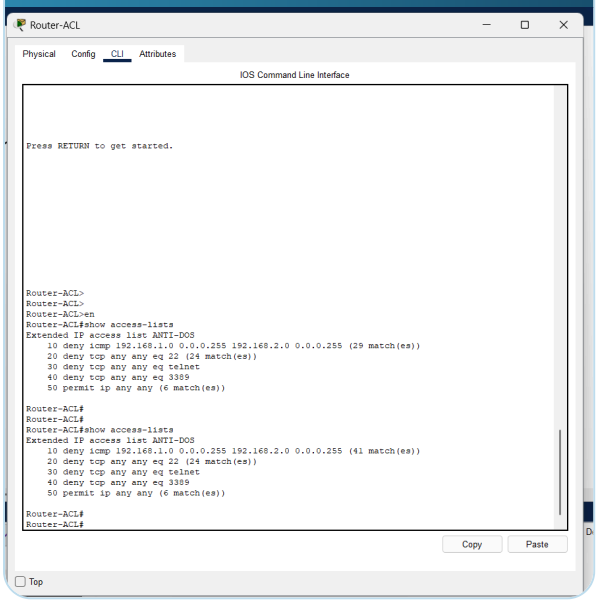
NOTE **Port Security Violation - Err-Disabled State**

After connecting an unauthorized device to interface `Fa0/2`, the switch detects a Port Security violation. The output of `show port-security interface Fa0/2` now shows that the port status changed to **Secure-shutdown**, the **Violation Count** increased to **1**, and the interface entered the **err-disabled** state. As a result, the connection is cut off completely and the MAC flooding attack is stopped successfully.

ACL Verification - Test Results

Test	Protocol / Port	Source -> Destination	ACL Counter	Result
ICMP Ping Test	ICMP	192.168.1.0/24 -> 192.168.2.0/24	29 -> 41 hits (up)	Blocked
SSH Connection	TCP Port 22	PC-Attack -> Protected Servers	-	Denied
Telnet Connection	TCP Port 23	PC-Attack -> Protected Servers	-	Denied
RDP Connection	TCP Port 3389	PC-Attack -> Protected Servers	-	Denied
Normal Traffic	Any	PC-Client1 -> Servers	-	Allowed

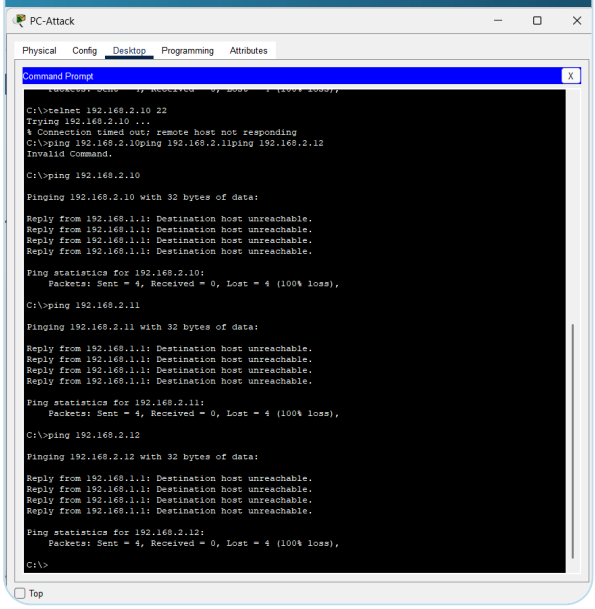
- ACL ICMP Deny Counter Increase



NOTE **ACL Verification - ICMP Deny Counter Increase**

Verification of the `ANTI-DOS` ACL on **Router-ACL**. The output of `show access-lists` shows that the match counter for the `deny icmp` rule increased from **29 hits** to **41 hits** after a ping test from PC-Attack. This confirms that the router detects and blocks ICMP traffic coming from the attacker subnet `192.168.1.0/24`.

- PC-Attack - All Connections Blocked



NOTE **PC-Attack - All Connections Blocked**

Attempts by **PC-Attack** to reach the protected servers fail completely. **ICMP ping** returns *Destination host unreachable*, and attempts to connect to **port 22** end with *connection timeout*. This proves that Router-ACL blocks malicious traffic before it can reach the protected servers, which is the main objective of the lab.

🎯 **Conclusion - All Lab Objectives Achieved**

All lab objectives were completed successfully: the `ANTI-DOS` ACL blocks risky traffic, Syslog records events as a substitute for ACL logging, and Port Security prevents MAC flooding attacks by placing the port into an `err-disabled` state when a violation occurs.